

| Question |     |      |  | Marking details                                                                                                                        | Marks Available |          |          |          |          |          |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                        | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 1        | (a) | (i)  |  | Microwaves                                                                                                                             | 1               |          |          | 1        |          |          |
|          |     | (ii) |  | Microwaves                                                                                                                             | 1               |          |          | 1        |          |          |
|          | (b) |      |  | Radio [waves] / X-rays / Gamma [rays]                                                                                                  | 1               |          |          | 1        |          |          |
|          | (c) |      |  | 3rd and 4th boxes ticked (1) + (1).<br>Additional boxes ticked –1 per box (minimum 0)                                                  | 2               |          |          | 2        |          |          |
|          | (d) | (i)  |  | Substitution: $170 \times 2$ (1) [Not: multiple selections unless correct final answer given]<br>Waves speed = $340 \text{ [m/s]}$ (1) | 1               | 1        |          | 2        | 2        |          |
|          |     | (ii) |  | Wavelength increases [or equiv] (1) [not: just stating the wavelengths]<br>..[so] speed increases (1)<br>NB Independent marks.         |                 | 2        |          | 2        |          |          |
|          |     |      |  | <b>Question 1 total</b>                                                                                                                | <b>6</b>        | <b>3</b> | <b>0</b> | <b>9</b> | <b>2</b> | <b>0</b> |